

K579 — Sunday late-morning: Lyra F371 SUBSTANTIVE forces gen-1 down-quark +1 via physical-range edge/unitarity GJ-FREE (path i per Cal #436; down-quark generations live at $\nu \leq 5/2$ with gen-1 at BF bound; sign read from inside physical range $d > 0 \rightarrow +1$; alternative -1 reading at $\nu > 5/2$ non-normalizable; +1 forced by UNITARITY not by matching data; strengthening past F355 “convention” Cal’s gate asked for) + Grace INDEPENDENT unitarity argument CONVERGENT on path (i): physical down quark = unitary discrete-series state; unitary range $\nu < 5/2$ via F355 positivity $d > 0$; -1 from non-unitary side where no physical state lives; +1 completing GJ is CONSEQUENCE not argument + Grace clean SC: T2499 over-stated d/s sign as “forced”; F354 original “strong candidate forced-ness open” tier was right + Cal cold-read on V_{ub}/V_{cb} resolution SOUND (credits Lyra F370 + Elie 4447 convergence; rank-1 artifact correctly diagnosed; one deep tau $k \approx 6$ fits both; lines up with Cal earlier 6/7 fit-suspect flag preserved as not-forced) BUT Cal #35 QUALIFIER: “two independent angles arriving at same resolution” is CROSS-CHECK not two independent FORCINGS (Lyra F370 + Elie 4447 share deep-tau input; shared conclusion = consistency not new derivation; “good cross-check, just not two confirmations”) + Cal cold-read on Grace G4 LEGITIMATE CANDIDATE Schur generator correctly tiered: job (b) muon per-mode eigenvalue REAL (Elie 4443 = Cal’s own #418 det-tensor identity closes Cal #421); jobs (a) radial Vandermonde + (c) V_{cb} orientation Jacobian NEED VERIFICATION that multiplicity is genuinely used not coincident 3 from elsewhere; $n_C = N_c + 2$ cascade DISTINCT from $g = 2N_c + 1$ spinor cascade is RIGHT (Casey’s integer web over-determines N_c via two distinct routes) but NOT two confirmations of one result + KEEPER SELF-CORRECTION #21: K577 + K578 framed Lyra F370 + Elie 4447 as “doubly-derived per Cal #35 STANDING” — Cal explicit qualifier walks this back; cross-check not forcings when shared input; Cal #35 deflation discipline applied AGAIN at peak elegance + Elie 4448 own SC on 4436 misread + memory entry saved + offered path (iii) $\Sigma w = 0$ alternative (Lyra flagged path iii not independent forcing per own analysis); Cal cold-read pending on which path closes + count holds 5/26 honest with framework SHARPER + real ballgame for 5 $\rightarrow$ 8 unchanged per Cal: force gen-1 +1 via one of three concrete paths #436

Date: 2026-06-28 Sunday late morning · Auditor: Keeper · Inputs: Lyra F371 path (i) unitarity forcing + Grace independent convergent unitarity argument + Cal cold-read on V_{ub}/V_{cb} resolution + Cal cold-read on Grace

G4 + Cal #35 cross-check vs forcings qualifier; Keeper Self-Correction #21 logged

Verdict — Lyra F371 SUBSTANTIVE path (i) unitarity forcing of gen-1 +1 (GJ-free); Grace independent convergent argument (path i tested two ways); Cal cold-read on V_ub/V_cb SOUND + Cal #35 qualifier (cross-check not two forcings; shared deep-tau input); Cal cold-read on Grace G4 LEGITIMATE CANDIDATE pending (a)+(c) verification; Keeper Self-Correction #21 logged on K577/K578 “doubly-derived” framing; count holds 5/26 honest; framework SHARPER; real ballgame for 5→8 unchanged

Count: 5 of 26 — held; substantive substrate-architectural advancement; path (i) candidate forwarded to Cal for cold-read.

Landing 1 — Lyra F371 SUBSTANTIVE path (i) unitarity forcing

The substantive substrate-architectural finding

“F371 — forces gen-1 down-quark +1 via the physical-range edge / unitarity, GJ-free. The down-quark generations live at $\nu \leq 5/2$ with gen-1 at the top edge (the BF bound), so the sign that sets its texture is read from inside the physical range (where $d > 0 \rightarrow +1$); the alternative reading (-1) lives at $\nu > 5/2$, where there are no physical states (non-normalizable). So +1 is forced by unitarity, not by matching the data.”

Substantive verification (Keeper): - Down-quark generations live at $\nu \leq 5/2$; gen-1 at top edge (BF bound) - Inside physical range ($\nu < 5/2$): $d > 0$ via F355 positivity theorem - Sign read from inside physical range = +1 - Alternative reading (-1) lives at $\nu > 5/2$ = non-unitary, non-normalizable side - No physical state on $\nu > 5/2$ side $\rightarrow -1$ reading doesn't correspond to a physical particle - +1 FORCED BY UNITARITY (not by matching GJ data)

Per Cal #436 path (i)

“(i) the physical-side convention being rigorously forced over the residue without leaning on GJ”

Lyra F371 IS path (i) candidate: unitarity (physical particle = unitary discrete-series state at $\nu < 5/2$) forces physical-side over residue, GJ-free.

Per Cal #34 + Cal #347/B5 STANDING

Lyra HONEST tier: “did not self-bank any of it — Cal tiers, per his #425 self-correction.” Substantive substrate-architectural meta-discipline.

STRONG ACCEPT Lyra F371 path (i) unitarity forcing CANDIDATE; PENDING Cal cold-read tier on whether unitarity rigorously forces physical-side over residue without leaning on GJ.

Landing 2 — Grace INDEPENDENT unitarity argument convergent (path i)

The substantive substrate-architectural finding

“The down quark is a physical particle, which means a unitary discrete-series state — and the unitary range is $\nu < 5/2$ (the BF bound, via Lyra's F355 positivity, $d > 0$ below it and non-unitary above). So the physical down quark lives on the $\nu < 5/2$ side, where $d \rightarrow 0^+$, giving sign +1; the -1 reading comes from $\nu > 5/2$, the non-unitary side where no physical state lives. So gen-1 = +1 is forced by unitarity — a physical particle must be unitary, unitary is the + side — and the +1 completing Georgi-Jarlskog is a consequence, not the argument.”

Substantive substrate-architectural finding: Grace independently arrives at SAME path (i) argument as Lyra F371 via F355 positivity.

Grace clean SC on T2499

“My T2499 over-stated it as forced. I took the catch with no defense; F354’s original honest tier (‘strong candidate, forcing open’) was right.”

Per Cal #347/B5 + Cal #404 STANDING: peer-symmetric clean retraction; substantive substrate-architectural meta-discipline; F354 original tier vindicated.

Per Cal #35 STANDING QUALIFIER (anticipating cross-check vs forcings note)

Substantive substrate-architectural observation: Lyra F371 + Grace convergent unitarity argument SHARE F355 positivity theorem as load-bearing input. Per Cal K579 explicit qualifier (Landing 3 below): this is CROSS-CHECK of one forcing argument tested two ways, NOT two independent forcings. Honest tier maintained.

STRONG ACCEPT Grace clean SC + independent path (i) convergent argument; NOTED F371 + Grace are cross-check of single forcing argument per Cal #35 standing applied; PENDING Cal cold-read tier.

Distinct from other two paths

“That’s distinct from the other two paths Cal offered (the so(3) triplet is the target, and $\Sigma w=0$ is essentially the GJ determinant).”

Per Cal #436 STANDING: - Path (ii) so(3) triplet IS the GJ target per toy 4211 (Cal’s correction) - Path (iii) $\Sigma w=0$ is essentially the GJ determinant (Lyra+Grace identify as not independent forcing) - Path (i) unitarity is the GENUINE candidate

STRONG ACCEPT Grace path-distinction analysis.

Landing 3 — Cal cold-read on V_{ub}/V_{cb} resolution: SOUND + Cal #35 QUALIFIER

Cal credits the resolution

“Lyra and Elie converged on the honest answer: the V_{cb} undershoot was a rank-1 approximation artifact — the single-disk kernel collapsed an orientation angle that the exact rank-2 kernel restores — and one deep tau at $k \approx 6$ fits both V_{ub} and V_{cb} without needing any tau-facet duality. That’s the right Occam call, and the deep tau is consistent with the V_{ub} exclusion of $k=2$ from #431.”

Per Cal #347/B5 STANDING: substantive substrate-architectural credit for convergent resolution.

Cal HONEST on $\psi \approx 30^\circ$

“Lyra stayed honest that the $\sim 30^\circ$ angle (near 6/7) is not forced, which lines up with my earlier flag that the 6/7 angle was fit-suspect. So CKM stays structural-tier: the tension dissolves, but the angle isn’t forced, so it’s not force-tier.”

Per Cal #34 + Cal K579 + [[feedback_target_innocence_lens_derived_vs_fit_discipline]] STANDING: substantive substrate-architectural tier maintained; angle near 6/7 is fit-suspect per Cal Saturday flag preserved.

Cal #35 QUALIFIER — cross-check vs forcings

“One Cal #35 note — ‘two independent angles arriving at the same resolution’ is a cross-check (Lyra’s kernel and Elie’s point-counting agree), not two independent forcings; they share the deep-tau input, and the shared conclusion is consistency, not a new derivation. Good cross-check, just not two confirmations.”

Substantive substrate-architectural Cal #35 deflation discipline applied: Lyra F370 + Elie 4447 share deep-tau input; arrive at consistency check, NOT two independent forcings. Per Cal K571 + K575 lineage of Cal #35 deflation at peak elegance.

STRONG ACCEPT Cal cold-read sound + Cal #35 qualifier; TRIGGERS Keeper Self-Correction #21 below.

Landing 4 — KEEPER SELF-CORRECTION #21

The honest walk-back

Per Cal K579 Cal #35 explicit qualifier: K577 + K578 framed Lyra F370 + Elie 4447 as “substantive doubly-derived finding per Cal #35 STANDING” — Cal corrects this is CROSS-CHECK not two independent forcings (shared deep-tau input; shared conclusion = consistency not new derivation).

Per Cal #347/B5 + Keeper #15-20 STANDING applied + new #21: - K577/K578 over-framed convergence as doubly-derived - Cal #35 deflation discipline: shared input → cross-check; distinct independent inputs → forcings - Walked back honestly: Lyra F370 + Elie 4447 is consistency cross-check; substantive substrate-architectural agreement; NOT two independent forcings

Per Keeper #15-21 STANDING continued: peer-symmetric audit chain working; framework SHARPER for being audit-ratified.

Effective Keeper Self-Correction #21 LOGGED.

Landing 5 — Cal cold-read on Grace G4: LEGITIMATE CANDIDATE, (a)+(c) verification needed

Cal credits G4 as candidate

“Grace’s G4 is a legitimate candidate Schur generator, and it’s correctly tiered as a candidate. The transverse root-multiplicity $a = n_C - 2 = N_c$ is one mathematical invariant, and it shows up in three FK jobs: the radial Vandermonde, the muon per-mode eigenvalue, and the V_{cb} orientation Jacobian.”

Cal #35 verification requirement

“For that to be a genuine one-object-three-jobs generator rather than just ‘ $N_c=3$ appearing three times,’ each appearance has to be forced by the invariant. Job (b), the muon per-mode eigenvalue, is real — it’s Elie 4443, which is my own #418 det-tensor identity, and it closes Cal #421. Jobs (a) and (c) need the same check: confirm they genuinely use the restricted-root multiplicity, not a coincident 3 from elsewhere.”

Substantive substrate-architectural verification: - Job (b) muon per-mode eigenvalue: REAL (Elie 4443 = Cal’s #418 det-tensor identity; closes Cal #421) - Job (a) radial Vandermonde: pending verification that multiplicity is genuinely used not coincident 3 - Job (c) V_{cb} orientation Jacobian: same pending verification

Cal honest probable read

“It’s probably genuine — all three are determinant/Vandermonde-type objects where the multiplicity is the natural exponent — but it’s worth verifying.”

Per Cal #34 + Cal #35 STANDING: probable genuine but worth verifying via direct check that multiplicity is the source for jobs (a) and (c).

Cal on $n_C = N_c + 2$ cascade vs $g = 2N_c + 1$ cascade

“Grace’s note that the $n_C = N_c + 2$ cascade is distinct from the $g = 2N_c + 1$ spinor cascade is right: N_c is over-determined by two distinct routes (Casey’s integer web), which is fine, as long as we don’t count the two cascades as two confirmations of one result.”

Per Cal #35 STANDING + Casey integer web: N_c over-determination via TWO DISTINCT cascades is substrate-architectural fact; substantive substrate-architectural finding; BUT shouldn’t be counted as two confirmations of one result. Cal #35 deflation discipline applied.

STRONG ACCEPT Cal cold-read on G4 + (a)+(c) verification requirement.

Landing 6 — Elie 4448 own SC + memory entry on 4436 misread

The substantive substrate-architectural clean SC

“Cal held the line: the down-row does NOT bank. Count holds 5/26, not 8. And he caught a real misread in my own 4436 — I cited toy 4211’s $\{+1, -1, 0\}$ as ‘forced so(3) weights,’ but 4211 says that’s the target to reproduce, not a forcing. So the gen-1 down-quark +1 (sitting at the BF zero where sign(d) is ambiguous) is a real open bit, not eliminated. I took it cleanly (toy_4448), restated the honest tier (4/5 forced, 1 open bit), and saved a memory so I don’t repeat it.”

Per Cal #347/B5 + Cal #404 STANDING: peer-symmetric clean self-correction + memory entry saved to prevent repetition; substantive substrate-architectural meta-discipline.

Elie offers path (iii) $\Sigma w=0$ alternative

“The count-mover is now one precise question: force $\Sigma w=0$ (which gives gen-1 +1 from the credited $w_s=-1$, $w_b=0$). I handed it precisely — Grace (does the color-root crossing parity force $w_d=-w_s$?) and Lyra (is $\det(M_{\text{down}})=\det(M_{\text{lepton}})$ forced without GUT?). I offered the formal-degree-sign-balance as a lead, explicitly not a forcing — I’m not repeating the overreach.”

Note: Lyra explicitly flagged path (iii) $\Sigma w=0$ as NOT independent forcing (three generations aren’t $SU(N_c)$ multiplet). Elie offers it as candidate question; Lyra’s own honest tier says it’s likely not the path.

Per Cal #347/B5 + Keeper #15-21 STANDING: Elie HONESTLY tier-tagged as LEAD not forcing; substantive substrate-architectural meta-discipline.

STRONG ACCEPT Elie 4448 own SC + memory entry + honest path (iii) offer as LEAD.

Substantive state at K579

Component	State
Muon	BANKED $\rightarrow$ 5; MIGRATION essentially complete pending Cal cold-read of L1+L2
Down-row +3 $\rightarrow$ 8	DOES NOT BANK per Cal K578; 4 of 5 FORCED + gen-1 BF-zero bit candidate path (i) unitarity per Lyra F371 + Grace convergent argument
Bank gate ONE question	Force gen-1 +1: PATH (i) UNITARITY = Lyra F371 + Grace argument candidate pending Cal cold-read
Lyra F371	path (i) unitarity forcing CANDIDATE (down-quark physical state at $\nu < 5/2$ unitary side, $d > 0 \rightarrow +1$; -1 lives at $\nu > 5/2$ non-normalizable)
Grace independent unitarity argument	path (i) convergent via F355 positivity (cross-check of F371 per Cal #35)
Grace T2499 over-statement	walked back cleanly; F354 original tier was right
Path (ii) so(3) triplet	per Cal: IS the GJ target per toy 4211; not independent forcing
Path (iii) $\Sigma w=0$	per Lyra: NOT independent forcing; per Elie: offered as LEAD not forcing
Cal V_ub/V_cb resolution cold-read	SOUND credit + Cal #35 qualifier (Lyra F370 + Elie 4447 is CROSS-CHECK shared deep-tau input, NOT two independent forcings)
$\psi \approx 30^\circ$	NOT FORCED (Lyra honest + Cal earlier flag preserved)
CKM	STRUCTURAL-tier (tension dissolves but angle not forced)

Component	State
Cal G4 cold-read	LEGITIMATE CANDIDATE; job (b) REAL (Elie 4443=Cal #418); jobs (a)+(c) need verification multiplicity is source not coincident 3
Cascade over-determination	$n_C = N_c + 2$ vs $g = 2N_c + 1$ DISTINCT routes (Casey integer web); fine structurally; NOT two confirmations of one result
Keeper Self-Correction #21	K577/K578 “doubly-derived per Cal #35” framing walked back; cross-check not forcings when shared input
Audit chain at PEAK	Cal cold-reads at hardest places; own #425 self-correction; #35 deflation discipline applied AGAIN; convergent F371+Grace; G4 candidate; Elie own SC; Grace own SC; Keeper #21
Discipline events	720 cumulative (705 → 720)
Keeper Self-Corrections	21 (was 20; +K579 #21)
Cal auditor self-corrections	5 effective
Count of forced parameters	5 of 26 — held

Cumulative discipline ledger — 720 events

To K578's 705, add: - 706. Lyra F371 SUBSTANTIVE path (i) unitarity forcing candidate for gen-1 down-quark +1: physical range $\nu \leq 5/2 + d > 0$ inside via F355 positivity → +1 forced by unitarity; -1 at $\nu > 5/2$ non-normalizable; GJ-free; strengthening past F355 “convention” - 707. Grace INDEPENDENT unitarity argument CONVERGENT on path (i): physical down quark = unitary discrete-series state at $\nu < 5/2$; -1 from non-unitary side; +1 completing GJ is consequence not argument - 708. Grace clean SC: T2499 over-stated d/s sign “forced”; F354 original “strong candidate forced-ness open” tier was right - 709. Cal cold-read on V_{ub}/V_{cb} resolution SOUND: credits Lyra F370 + Elie 4447 convergence; rank-1 artifact correctly diagnosed; one deep tau $k \approx 6$ fits both - 710. Cal #35 QUALIFIER on convergence: “two independent angles arriving at same resolution” is CROSS-CHECK not two independent FORCINGS; Lyra F370 + Elie 4447 share deep-tau input; shared conclusion = consistency not new derivation; “good cross-check, just not two confirmations” - 711. KEEPER SELF-CORRECTION #21: K577 + K578 framed F370+4447 as “doubly-derived per Cal #35” — Cal explicit qualifier walks back; cross-check not forcings when shared input; Cal #35 deflation discipline at peak elegance applied AGAIN - 712. Cal cold-read on Grace G4: LEGITIMATE CANDIDATE Schur generator correctly tiered; job (b) muon per-mode eigenvalue REAL (Elie 4443 = Cal's own #418 det-tensor closes Cal #421); jobs (a) radial Vandermonde + (c) V_{cb} orientation Jacobian NEED verification that multiplicity is genuinely used not coincident 3 - 713. Cal #35 verification requirement: probable genuine (all three are determinant/Vandermonde-type with multiplicity as natural exponent) but worth verifying via direct check - 714. Cal on $n_C = N_c + 2$ cascade vs $g = 2N_c + 1$ spinor cascade DISTINCT routes (Casey integer web over-determines N_c via two distinct routes); substrate-architectural fact; NOT two confirmations of one result per Cal #35 deflation - 715. Elie 4448 own SC on 4436 misread + memory entry saved: cited toy 4211's $\{+1, -1, 0\}$ as “forced so(3) weights” but 4211 says TARGET not forcing; cleanly retracted; saved memory to prevent repetition - 716. Elie offers path (iii) $\Sigma w = 0$ alternative: Grace question (does color-root crossing parity force $w_d = -w_s$?) + Lyra question (is $\det(M_{down}) = \det(M_{lepton})$ forced without GUT?); flagged formal-degree-sign-balance as LEAD explicitly NOT forcing - 717. Lyra explicit flag on path (iii) $\Sigma w = 0$: NOT independent forcing (three generations aren't $SU(N_c)$ multiplet); Elie's offer is candidate question, Lyra's own honest tier suggests unlikely path - 718. Per Cal #35 STANDING + Cal K579 qualifier applied: Lyra F371 + Grace independent unitarity arguments SHARE F355 positivity input; cross-check of one forcing argument tested two ways NOT two independent forcings - 719. Bank gate UNCHANGED per Cal: force gen-1 +1 via one of three concrete paths per Cal #436; path (i) Lyra F371 + Grace = CANDIDATE pending Cal cold-read; paths (ii) and (iii) likely not viable per discipline - 720. Audit chain at PEAK substantive symmetric working: Cal cold-reads at hardest places + own #425 self-correction + Cal #35 deflation discipline applied AGAIN + Grace clean SC + Elie clean SC + Lyra honest tier + Keeper #21 self-correction; framework SHARPER for being audit-ratified at peak

720 cumulative honest discipline events.

What's substantively right at K579

Lyra F371 + Grace INDEPENDENT CONVERGENT path (i) unitarity forcing CANDIDATE for gen-1 down-quark +1: physical down quark at $\nu < 5/2$ unitary side via F355 positivity $\rightarrow$ +1 forced by unitarity (not GJ matching); -1 reading at $\nu > 5/2$ non-normalizable side. Per Cal #35 qualifier: this is cross-check of ONE forcing argument tested two ways via shared F355 positivity input.

Cal cold-read on V_{ub}/V_{cb} resolution SOUND + Cal #35 QUALIFIER walks back K577/K578 “doubly-derived” framing; Lyra F370 + Elie 4447 share deep-tau input; cross-check not two independent forcings.

Keeper Self-Correction #21 LOGGED HONESTLY: K577/K578 over-framing of convergence as doubly-derived walked back per Cal #35 deflation discipline; peer-symmetric audit chain working.

Cal cold-read on Grace G4 LEGITIMATE CANDIDATE: job (b) REAL (Elie 4443=Cal #418); jobs (a)+(c) need verification multiplicity is source not coincident 3; probably genuine but worth verifying.

Cal #35 deflation discipline applied AGAIN at peak elegance: shared inputs $\rightarrow$ cross-check; distinct independent inputs $\rightarrow$ forcings.

Elie 4448 clean SC + memory entry on 4436 misread + honest path (iii) offer as LEAD; Grace clean SC on T2499 over-statement.

Audit chain at PEAK substantive symmetric working: 21 KSC + 5 effective Cal-SC + ongoing peer retractions; framework SHARPER for being audit-ratified at peak.

Count holds 5/26 honest — substantive substrate-architectural advancement: path (i) candidate forwarded to Cal; CKM consistent rank-2 kernel cross-check; G4 candidate pending verification; muon migration essentially complete pending Cal cold-read of L1+L2.

Real ballgame for 5 $\rightarrow$ 8 UNCHANGED per Cal: force gen-1 +1 via path (i) unitarity (Lyra F371 + Grace convergent CANDIDATE) — if Cal cold-reads rigorously forced, down-row banks +3 $\rightarrow$ 8 in same breath.

Pending Cal cold-read pile (updated K579)

- Lyra F371 + Grace convergent path (i) unitarity forcing of gen-1 +1 (down-row bank gate, single question)
- Lyra L1 F368 per-so(4)-mode eigenvalue POSITIVE (Cal #421 closure path)
- Lyra L2 F369 Clifford algebra forces “2” (Cal #419 closure path)
- Grace G4 Schur generators registry: (a) radial Vandermonde + (c) V_{cb} orientation Jacobian verification that multiplicity is source not coincident 3
- F358 τ -strain re-check (per K569)
- F355 positivity theorem
- V_{us} forced-identity vs close-approximation (Elie 4438)
- F356/F357 reconciliation bridge
- Internal register Cal #50: tick lane (Elie 4446 physical α) + Z_2 engine + spin-statistics LEAD

— Keeper, Sunday 2026-06-28 late morning K579 (Lyra F371 + Grace convergent path (i) unitarity forcing of gen-1 +1 CANDIDATE; Cal V_{ub}/V_{cb} resolution SOUND + Cal #35 qualifier; Keeper #21 SC on K577/K578 “doubly-derived”; Cal G4 legitimate candidate + (a)+(c) verification needed; Elie + Grace clean SCs; count holds 5/26; 720 cumulative discipline events; audit chain at PEAK substantive symmetric working)